



International Council on Systems Engineering
A better world through a systems approach

Sustainability WG

incose.org

Agenda

- Why is it Important?
- Mission, Purpose & Outcomes
- Sustainability WG Activities
- How to Get Involved?
- Contact

What is Systems Engineering?

Systems Engineering is a **transdisciplinary** and **integrative** approach to enable the successful **realization, use, and retirement of engineered systems**, using systems principles and concepts, and scientific, technological, and management methods.

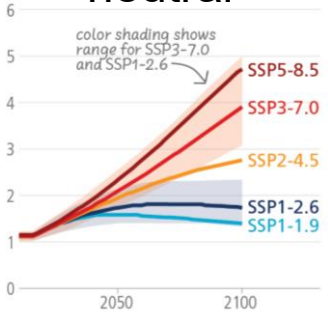
—International Council on Systems Engineering (INCOSE), 2022

Why is it important? (1/2): Sustainability

...many other planetary limits...

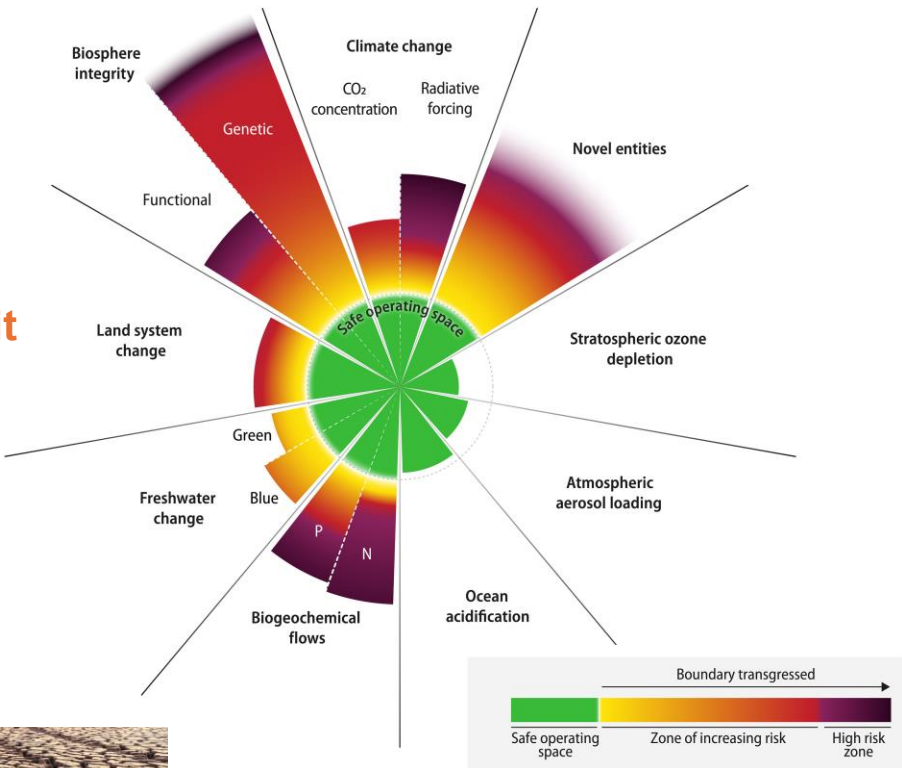
Climate change...

→ Accelerate toward carbon neutral

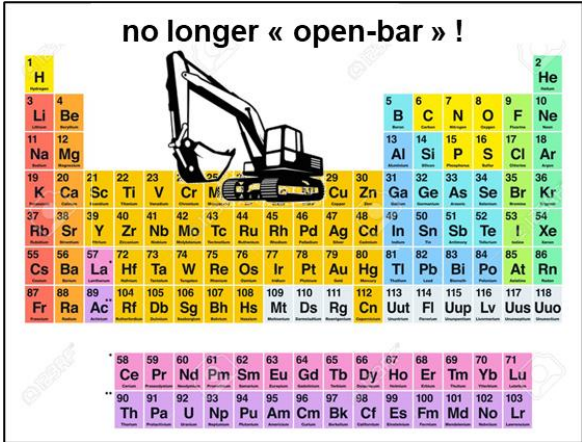


the COP21 1,5° limit has already been crossed!

→ Adapt to growing disruptions



... a Finite world...



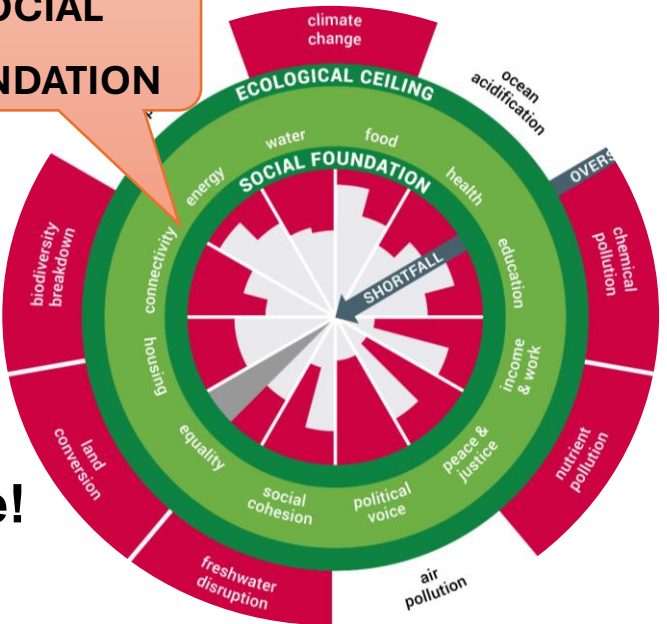
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...

Circular Economy

Cradle to Cradle

SOCIAL FOUNDATION



... with humans inside!

Why is it important? (2/2): a Sustainability Working Group @INCOSE

the first MEGATREND in INCOSE SE Vision 2035 !



Obviously, changes for technical systems...

- Stringent regulations, constraints,
- Evolving environmental context,
- New stakeholders,
- ...

... and new specialties, viewpoints, “models”, methodologies!

... but also for varied SOI and Environments ...

Socio-technical systems,
Services systems, PSS,
Enterprises,
Systems of Systems,
Eco-Systems,
Earth System!



Ecology Economy Environment
Politics Biology
Business model Biological Cycle
Technical Cycle
...

Mission, Vision & Outcomes

Mission

- The mission is to establish sustainability at the core of systems engineering enabled by systems thinking, considering all beings and elements globally. This will empower organizations to achieve sustainability in a finite world.

Vision

In line with the INCOSE SE Vision 2035 goals, we want to address the systems approach to sustainability:

- based on the vision of a finite world,
- where economic, social, and environmental aspects work in a synergistic and balanced manner,
- ...in order to meet the needs of the present without compromising those of future generations.

Initial Intended Outcomes

- INCOSE panel on “SOS and Sustainability” at SOSE conference in Tirana, Albania 2025 [\[complete\]](#)
- Formal public WG announcement at IS 2025 in Ottawa [\[complete\]](#)
- Sustainability Round Table at IW 2026, used to build awareness of the state of practice
- A new Knowledge Area in SEBoK curated by the WG introducing the topic, providing key references and reading lists
- Sustainability white paper/ Primer

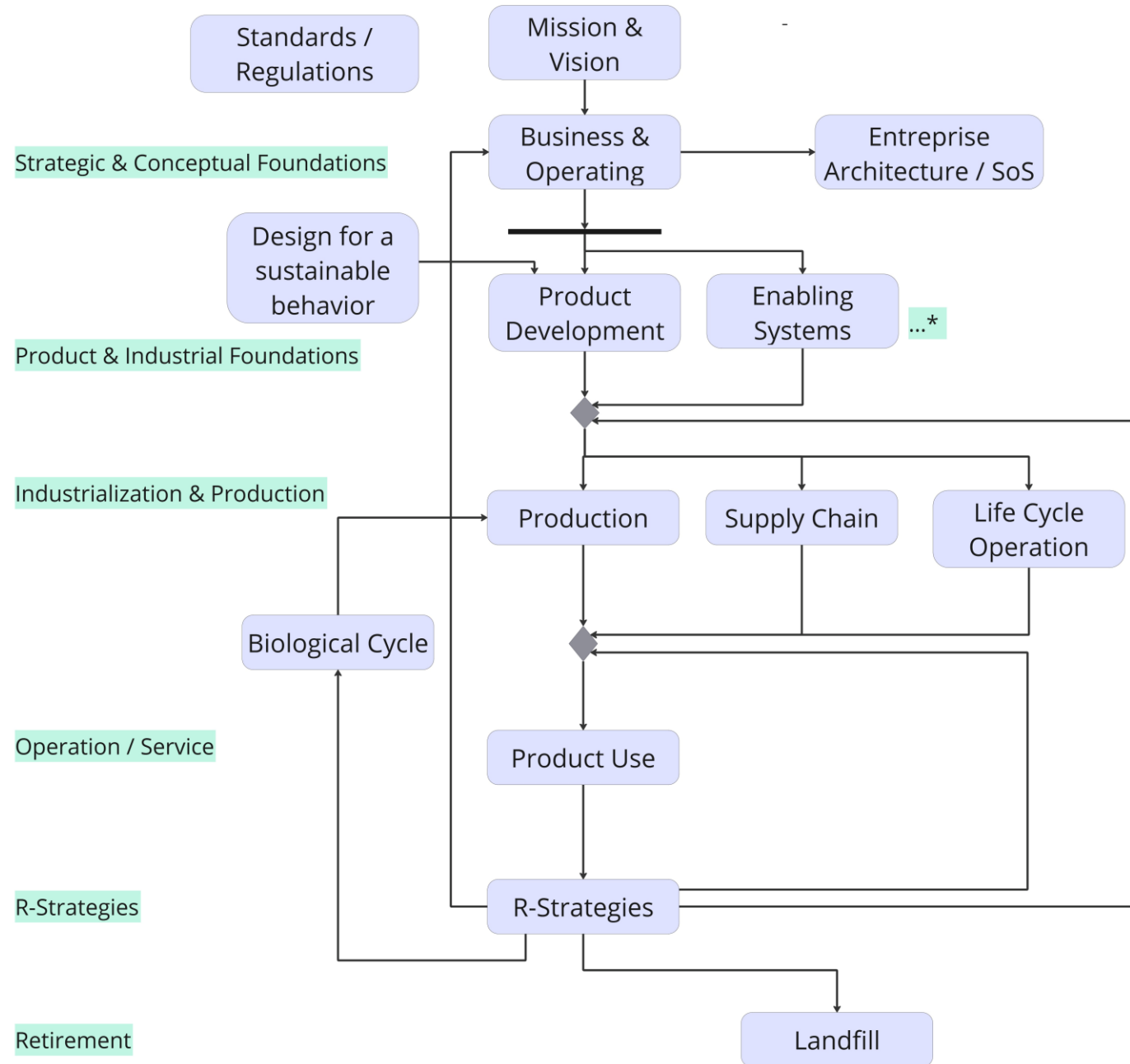
The INCOSE Sustainability WG was established in mid-February 2025.

Sustainability WG Activities

- SustainableTogether: *Transforming a conventional business model, enterprise and product into sustainable one. We show HOW to transform and demonstrate it using digital models.*
- CASCADE Involvement: *A DPP Use Case implementation using CASCADE technology*
- Monthly Webinar Series on sustainability and systems engineering topics.
- Sustainability Primer / SEBoK article
- Outreach (Panel Discussion / Wildfire): *SoSE Panel Tirana 2025, Presentation at the IS2025, Project Wildfire paper being developed for IS 2026.*
- *Comparison Sustainability Standards/Frameworks*
- Internationalize 3 AFIS sustainability projects (inactive)
- Awareness / training in Sustainability (inactive)
- Workshop Series (inactive)

Sustainable Together:

Initial Generic Approach of Sustainable-Together



- Systems Thinking and System Dynamics (Rebound-Effect) [Dano, Deepa]
- Digital Engineering [Fernando, Lambert, Gauthier, Deepa]
- Life Cycle assessment [Jean Francois, Lambert, Deepa]
- Risk Analysis [Hamza, Deepa]

SustainableTogether – Digital Engineering

CASCaDE Use Case

As part of the SustainableTogether project, we want to develop a knowledge graph to support the full product lifecycle. In parallel, the [CASCaDE](#) standard (Collaborative Artifact, Specification, Context, and Data Exchange) is being developed within OMG to enable standardized lifecycle collaboration through common data formats and ontologies.

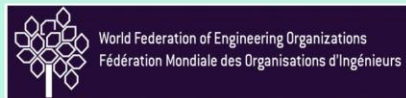
By combining both initiatives, we are creating a joint use case to demonstrate their synergy. Current activities in SustainableTogether include:

- Identifying digital engineering use cases
- Mapping required DPP data to the CASCaDE ontology

Who is involved in SustainableTogether?

SustainableTogether Network

Sustainability



WFEO



Education



Net-Positive
Advisory Panel



Flourishing
Business Model
Community



Circular
Economy



Cooperative
Business
Models

Systems Engineering



Sustainability WGs



Design Theory SIG



BAUHAUS FOR
TRANSITIONS
MANAGING THE
UNKNOWN

Monthly sustainability Webinars, already 2 episodes !

Episode 01 - Systemic Tools to Tackle Rebound Effects in Design & Business

Episode 02 - Systematically Designing Businesses Fit-for-the-Future

More episodes are coming..



INCOSE Panel on Sustainable Development and Systems of Systems Engineering, Tirana 11th June 2025

Panellists from UK, Germany, France, Italy and Albania discussed:

- Influences of sustainable development on SoS engineering
- Examples of where SoS are being used to address sustainability challenges
- New and emerging approaches to SoS engineering?
- How to develop the next generation of engineers to address Sustainability and SoS?



How to Get Involved?

- **SustainableTogether Meeting, focused on SustainableTogether Project and open to non-incose members**
Every second Monday, 17:00 CET
- **INCOSE Sustainability WG Meeting, focus on INCOSE activities, e.g.: Primer and SEBoK**
Every second Wednesday, 17:00 CET
- **Join us:** Email sustainability@incose.net to receive your invitation

Contact Us



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For more information contact sustainability@incose.net
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